Water Resources Research

RESEARCH ARTICLE

Target-specific diagnostics improve screening of river-temperature extremes, but transfer remains uncertain

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Key Points:

- Target-specific diagnostics improve H30 screening beyond daily diagnostics at matched reconstruction and learning capacity
- Extra peak-window summaries do not establish a stable practical gain over strong direct H30 risk references
- Calibration transfer, local sample size and the cost of unknown answers limit the interpretation of screening gains

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Abstract River-temperature archives support assessments of sustained thermal extremes, yet record completeness does not determine whether those extremes remain recoverable. Here, we use 6,170 complete reference station-years, paired gap experiments and frozen tests on new European station histories to examine the annual hottest consecutive 30-day mean (H30). We separate the effects of missing-window support, diagnostic target and calibration context. For equal gap length, moving a 30-day gap into the climatological peak raises climatology-based H30 absolute error by 0.62 °C; extending a fully hidden peak also increases error. H30 diagnostics improve screening relative to daily diagnostics: local recalibration on Swiss records yields 9.4 additional retention points [6.1, 13.1]. Further peak-window summaries do not establish a stable five-point advantage over strong direct H30 uncertainty references, including matched failure-probability learners and a developed conditional distribution. The frozen window rule retains 56.1% of 669 Swiss cases but incurs 8.8% failures against a 5% calibration target. Local recalibration lowers observed risk to 4.1%, although seven test stations cannot validate a population guarantee. Numerical screening and hot-year classification also rank policies differently; inexpensive unknown answers can outperform the original screen. These findings support target-specific assessment of thermal archives while identifying calibration transport, conditional distribution quality and intended use as separate limits on record fitness.

Plain Language Summary A temperature record can contain many observations and still miss the weeks needed to estimate a river's annual thermal extreme. We test which incomplete years can support the hottest 30-day mean. Diagnostics aimed at that statistic retain more usable estimates than diagnostics of daily reconstruction error. Adding a more detailed description of the uncertain hottest period gives no consistent further advantage over strong direct diagnostics. A screening rule calibrated on existing rivers also exceeds its target error rate when transferred unchanged to new Swiss records. Local examples improve the observed result, but too few independent rivers are available to establish a reliable population error limit. Finally, reporting an accurate temperature, identifying a hot year and withholding an answer have different consequences. Archive assessment therefore needs to specify both the quantity being reported and the evidence supporting its use.

1. Introduction

River-temperature records describe thermal exposure and climatic extremes and support assessments of aquatic habitat (Caissie, 2006). Many of these applications depend on sustained warm conditions rather than individual daily temperatures. The annual hottest consecutive 30-day mean, H30, captures one such property. Recovering it from an incomplete record requires information about the warmest part of the year: high annual coverage can coexist with an unobserved thermal peak, whereas a substantial cool-season gap can leave that peak well constrained. The scientific question is how the location and persistence of missing observations affect the information available for a specified thermal statistic.

Previous studies establish that missing-data timing affects annual stream-temperature analyses (Johnson et al., 2021) and that seasonal information can support thermal-peak reconstruction from heterogeneous records (Beaufort et al., 2022). This work provides a basis for assessing incomplete archives beyond a single coverage threshold. It leaves a further comparison to resolve: whether diagnosing error in the intended statistic adds useful information beyond daily reconstruction diagnostics, and whether explicitly representing its uncertain peak window adds a further benefit. These are distinct questions. A comparison that changes both the diagnostic target and its window representation cannot attribute improvement to either

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Table 1: The three scientific claims and the comparisons that support their scope. The primary use is delivery of an archived H30 magnitude; hot-year decisions are secondary sensitivity analyses.

Claim	Comparison and population	Evidence and limit
Target diagnostics improve H30 assessment	Daily versus H30 disagreement; fixed point, matched learner; source admission and original Swiss new-station test	Retention, accepted failures and component intervals; target awareness is distinguished from a necessary explicit window feature
Additional window representation has not established practical superiority	Direct H30 uncertainty versus added windows; matched quantile and failure-probability objectives	Five-point superiority and two-point equivalence are distinct; opened-data supplements do not create new confirmation
Calibration transport and use value require separate evidence	Original F0/F1/F2, paired information removal and magnitude screening versus direct cost decisions	Original candidate denominators, group failures and absolute losses; lower point risk is not certification or a general management benefit

component.

The distinction matters because several diagnostics can encode the same seasonal support. Calendar position, historical peak timing and disagreement among reconstructed H30 estimates may already describe much of the information carried by an explicit overlap variable. Furthermore, the true peak window is unavailable when a record is assessed. Its location can be used to diagnose reconstruction errors in a complete reference record, but a deployable rule must depend on visible observations and permitted history. Matched comparisons are therefore needed to separate the value of the target statistic from the value of a particular representation of its support.

A useful diagnostic must also retain its meaning across rivers and uses. Selective prediction provides established tools for exchanging the number of returned estimates against their error risk (Geifman & El-Yaniv, 2017; Angelopoulos et al., 2025). However, a threshold calibrated under one mix of reconstruction inputs and river conditions need not attain the same risk after transfer. Stream-temperature models with similar accuracy within their development environment can respond differently to environmental change (Topp et al., 2023). A further distinction arises between numerical accuracy and classification: an H30 estimate can exceed a numerical tolerance while retaining the correct hot-year label, or cross a classification threshold despite a small numerical error.

We combine complete reference records, paired missing-data experiments and frozen new-station tests to address three questions. First, how do gap position, duration and persistent reconstruction error propagate into H30? Second, how much screening information comes from the diagnostic target, and how much from additional peak-window summaries? Third, which calibration and use conditions preserve the value of that screening? The primary use is delivery of an archived H30 magnitude; hot-year classification examines a secondary use rather than an ecological hazard standard. Table 1 links the resulting claims to their comparisons. The original frozen experiment retains its hypotheses (Table 2); matched learning-objective, stronger-reference and failure analyses subsequently conducted on opened outcomes are identified as supplementary.

2. Data and study design

2.1. Comparisons and information available at use

The design separates three changes that can otherwise produce an apparent screening gain: changing the reconstruction, changing the diagnostic target and adding a representation of the target window. Paired gap experiments hold the reference station-year fixed while varying missing-data geometry. Diagnostic comparisons hold the delivered H30 estimate, learning capacity and calibration resources fixed. Transfer tests then change the station histories while preserving specified model and threshold choices. This sequence distinguishes error propagation, component attribution and deployment performance.

We use three evidence populations. Constructed gaps identify the consequences of timing and duration;

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Table 2: Original questions fixed before the original new-station experiment. Post-review supplements do not change these hypotheses. Failure means absolute H30 error above 0.5 °C. The empirical calibration target is 5%, not a guarantee of deployment risk.

	Hypothesis	Required comparison	Falsifier
H1	Target-window information helps beyond missing-day and flexible seasonal summaries	Paired timing/length controls; matched small and flexible models	No reproducible increment over seasonal comparators; an explicit-feature ablation alone is insufficient
H2	Uncertain support, reliability and competition give a practical admission gain	Same learner, point, labels and calibration; generic H30, direct uncertainty and certificate references	No independent gain beyond five retention points at the common risk target; equivalence uses a separate two-point margin
H3	Admission improves a locally hot-year decision	Identical original pool; reconstruction, screening, classification and charged abstention	No improvement over reconstructing all under declared positive abstention costs; lost information cannot be hidden

a broader mixture of short, long and empirical gaps evaluates admission; recorded-calendar replay examines historically observed missing-data geometries. Complete reference records provide evaluation values in these experiments. Independently backed natural gaps require a separate reference qualification and are not supplied by replay. Original frozen tests and later analyses of opened outcomes retain distinct status throughout.

2.2. Reference records and independent roles

The source cohort contains 581 provider-station identities and 6,170 complete station-years from 10 providers. Its 193 network labels form 181 connected components when shared target or donor records are accounted for. We use these components as the sampling unit to avoid treating reused records as independent evidence. The primary measurement scope requires provider-documented daily means; uncertain statistics and project-aggregated means are separate sensitivities. Within this scope, the controlled experiment includes 312,825 masks on 4,720 station-years in 144 components. A complete reference has a finite value on every calendar day. Completeness alone establishes neither sensor accuracy nor the absence of provider-estimated values.

New station histories were recruited from the mountain-river resource of van Hamel and Brunner (2024a, 2024b). Because its extreme-value subset cannot define a continuous H30 reference, we reacquired continuous temperatures from Swiss FOEN and French Hub'Eau. We audited all previously used target and donor identities using aliases, coordinates and available catchment connections. Candidate components were randomly assigned development, calibration and test roles before acquisition of daily values. Subsequent completeness and prior-input checks reduced eligibility while retaining those roles.

The resulting cohort contains 53 stations: 21 for development, 13 for calibration and 19 for testing. The test set includes 7 Swiss documented-mean and 12 French project-aggregated records. French daily aggregation requires at least four observations spanning eighteen local-time hours and includes explicitly unqualified source observations. Those records therefore remain outside the primary documented-mean scope. Eligibility requires four complete reference years, leaving at least three permitted complete training years for an evaluated year. Supporting information (SI) documents acquisition and exclusions.

The European source and target records are station-disjoint after audit, although incomplete historical catchment polygons limit the assessment of hydrological connections. A further transfer uses USGS-only source models, removing shared source–target river-flow connections while retaining possible climatic dependence and environmental shift. Descriptive process strata use glacier influence first and a majority-snow threshold of 50%. Regulation and groundwater classes remain unverified. These descriptive labels do not enter the frozen predictors or thresholds.

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2.3. Gap timing, duration and recorded calendars

To isolate missing-data geometry, we pair 30-, 60- and 90-day gaps at feasible zero, partial and full overlap with the training-only climatological peak window. Four start realizations are attempted in each station-year cell. Further controls hold the number of missing warm-season days fixed or distribute that budget among contiguous and fragmented gaps. Infeasible combinations remain empty. Across measurement classes, the experiment yields 406,015 retrospective and 290,322 earlier-years-only cases. Replicated masks resolve placement effects within records; they do not add independent rivers.

Admission requires a wider population because a long-gap-only design omits records that a simple duration rule could accept. We therefore use lengths of 14, 30, 60, 90, 120, 180 and 270 days, one feasible climatological-overlap class per length, and up to five transplanted empirical geometries. This yields 70,011 retrospective cases, comprising 40,232 single-run and 29,779 empirical geometries, and 46,510 earlier-years-only cases. The mixture defines a screening assay rather than an estimate of natural outage prevalence.

Recorded-calendar replay transfers genuinely incomplete historical shapes onto complete reference years. It does not observe the original incomplete years' unknown values and excludes calendars with no missing or no observed days. Equal-duration event masks fall before, during and after the same longest thermal episode. Project episodes exceed a training-only seasonal 95th percentile for at least three days; published events use a separate two-day definition. Compound heat/low-flow labels use training-month discharge quantiles. Discharge missing on required gap-event days remains unknown.

We cross timing with removal of none, half or all initially usable neighbours, recomputing local reconstruction inputs but leaving admission heads and thresholds unchanged. Fractional removal rounds upward, so a one-neighbour case loses it at both nonzero levels. The new European bundles are single-station: their cofailure contrast would be degenerate. The nondegenerate test uses held-out opened-source multi-station components. Event-dependent masking is informative-missingness stress, not identification of real instrument-failure causes.

3. Target estimation, screening and validation

3.1. The thermal statistic and its reconstruction

For a complete recorded year y,

$$H_{30}(y) = \max_{w \in \mathcal{W}_{30}} \frac{1}{30} \sum_{t \in w} y_t, \tag{1}$$

where $\mathcal{W}_{30}$ contains consecutive windows within that calendar year. Let M denote missing days and S a maximizing true window. A climatological window S_c uses only permitted history. Its overlap $|M \cap S_c|/30$ is available at decision time; true overlap $|M \cap S|/30$ is an evaluation diagnostic, never a screening feature.

The archive-use benchmark asks whether to return an annual magnitude within 0.5 °C. Its 5% selected-risk cap concerns errors exceeding that tolerance under the declared group-sampling scheme. A five-percentage-point retention margin asks for five extra annual magnitudes per hundred original candidates at the common risk criterion. These parameters make the information-return tradeoff explicit; they are experimental benchmarks, not biological thresholds or validated agency service levels.

We fit day-of-year and smooth climatologies, interpolation, neighbour ridge and air-temperature ridge. The common point follows a fixed information priority: usable neighbour prediction, available air prediction, interpolation if the actual longest run is at most 30 days, then smooth and day-of-year climatology. Every hidden day must receive a finite value; observations stay unchanged. This is an information-priority policy, not selection on hidden error. Individual reconstructors and a certificate choosing its own method are separate comparisons.

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Writing $\tilde{y}$ for a completed year and $\hat{S}$ for its maximizing window gives

$$H_{30}(\tilde{y}) - H_{30}(y) = B + F, \tag{2}$$

$$B = \frac{1}{30}\sum_{t\in S}(\tilde{y}_t - y_t), \tag{3}$$

$$F = \bar{\tilde{y}}_{\hat{S}} - \bar{\tilde{y}}_{S} \geq 0. \tag{4}$$

B is error carried by the true window, with zero contribution from observed days. F is a *window-switch increment*. Another observed window can replace an underestimated true peak, making F positive while final error stays negative. It is not generally a false-hot-peak channel. A stratified absolute-error association between aligned and misaligned windows is not identified with F.

We cross hidden-residual permutation with removal of their hidden mean, including the interaction. Both require truth. A separate deployable correction uses visible flanks and historical residual information with shrinkage. Oracle centering benefits are not operational improvements or evidence of unconditional physical cold bias: conditioning on a true maximum can favor negative error there.

3.2. Separating the diagnostic target from window representation

Leave-training-year-out prediction residuals supply contiguous blocks for 128 conditional completed trajectories around the policy reconstruction. Blocks are nominally fourteen days and never bridge missing residual runs. If none exists for a shorter interpolation gap, a complete block as long as its longest run is permitted; long gaps never fall back to independent singletons. Observations stay fixed. Trajectories yield H30 spread/error-scale summaries, probabilistic hidden support, peak-location dispersion, competition and interactions. Bands are raw scenarios, not confidence or risk guarantees.

The uncertain-window score adds window summaries to a generic model containing gap, season, history, donor and reconstructed-H30 disagreement features, plus direct H30 uncertainty. Primary comparisons share quantile boosting, capacity, labels, point and calibration resources. Alternatives include direct H30 uncertainty; actual out-of-fold daily reconstruction scale; complete historical/observed diagnostics; mask-matched historical H30 error; seasonal splines; and small logistic models with and without overlap. Replacing H30 disagreement with daily prediction disagreement tests target awareness, not just explicit overlap. Removing competition, window spread and donor features tests the components.

The original score learns the 0.9 quantile of absolute H30 error. An additional four-cell comparison crosses that objective with direct learning of $P(|H_{30} - \hat{H}_{30}| > 0.5 \mid \mathcal{J})$, each using direct H30 diagnostics alone or with added window summaries. All cells share the reconstruction, component roles, boosting depth, iterations, learning rate and leaf size. Complete risk–retention curves, risk at fixed retention and full learning/calibration refits evaluate whether apparent window gains depend on the learning objective. These comparisons use opened data.

A common-point certificate proxy differs from an aligned head for the current policy. A whole-system certificate chooses its own reconstruction and calibrates that point's error; it is not a feature-only comparison. Unsupported scores do not remove candidates. Air2stream-3/8 (Piccolroaz et al., 2016) and an old-source-trained univariate CSDI reference (Tashiro et al., 2021) test reconstruction sensitivity on identical masks, with availability and matched-subset policy errors. CSDI has no fully prospective claim. Air2stream forcing calibration and normalization exclude the target year and use earlier years prospectively. Extra meteorology, discharge or neighbours are information increments, not architectural gains.

3.3. Conditional H30 distributions

A further probability-reference budget compares two air-conditioned CSDI widths using old group-disjoint data only. Fitting has 3,469 years, while thirty old validation groups each provide five predeclared peak, boundary and fragmented masks. Early stopping uses masked-water validation loss; configuration selection uses H30 CRPS in degrees Celsius. Source normalization excludes the target year, and observed water

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remains fixed. We report compute and 8/16/32-sample sensitivity. Distribution samples are shared across reconstruction points so direct failure probability is computed for each delivered point before testing added windows. Calibration and use decisions retain their independent roles and original pools.

3.4. Calibration transport and the independent unit

Source comparisons use five outer folds with disjoint fitting, calibration and testing components in approximately 60/20/20 proportions, repeated for three seeds. Candidate thresholds are fixed from training scores. The empirical selector uses a calibration component bootstrap and is heuristic. A zero-failure bootstrap can return zero uncertainty without assigning probability to an unseen failure type; more repetitions do not solve that problem.

For a formal alternative, let A_g be within-component mean admission and B_g mean admission times failure. Test $X_g = B_g + 0.05(1 - A_g)$. A Chernoff–KL upper bound on $E[X_g]$ at most 0.05 implies $E[B_g]/E[A_g] \leq 0.05$, with positive denominator. Bonferroni adjustment permits searching training-fixed candidates without monotonic-risk assumptions. This requires independent, deployment-representative components and a predictor independent of calibration. It concerns a group-expectation ratio, not every subgroup, arbitrary archive case risk or unconditional transfer. Even with zero failures and full admission, one threshold needs 59 independent groups; 21 candidates need 118. Our target calibration cohorts cannot establish this guarantee.

Deployment separates F0, unchanged source model and threshold; F1, the same model with target calibration; and F2, new-development fitting with target calibration. Provider-excluded and USGS-only transfers are further checks. Local historical reconstruction fitting is allowed; hidden target-year labels never select a method. Predictive choices are frozen before target calibration errors; thresholds are locked before test errors.

Earlier-years-only local reconstruction is not a complete forward-in-calendar-time pipeline. Our forward experiment also fits and calibrates the global head through 2018, evaluates 2019–2022, and computes local climatology, donor relationships and residual diagnostics from earlier calendar years. This is a retrospective forward hindcast using current-vintage observations and reanalysis, not a real-time-vintage or prospective field trial; historical publication and revision vintages are unavailable. Its smaller pool is separate. Whole-procedure uncertainty refits learning and calibration in 1,000 component resamples per protocol, unlike intervals conditional on fitted procedures. Learning curves use repeated subsets. Power projections retain training/calibration variability and do not establish common-risk feasibility under shift.

3.5. Numerical delivery, hot-year decisions and unknown records

The primary label asks whether H30 exceeds the 90th percentile of permitted training-year H30. This finite-history local anomaly proxy is not a verified ecological or regulatory standard. Fixed 18 and 20 °C labels are sensitivities. False-hot, missed-hot, correct and unknown outcomes share the original pool, with separate hot/cool refusal, threshold proximity, coverage and regional-frequency distortion.

With false-hot cost one, loss is $(c_{FN}FN + FP + c_A U)/N$. Declared missed-hot costs are 1, 5 and 10; unknown-answer costs are 0, 0.01, 0.025, 0.05, 0.1 and 0.25. Refused records remain unknown or interval-valued, not automatically repaired. These are not measured agency preferences; intervals across the grid are descriptive, not simultaneous guarantees.

A use-matched rule estimates $p = P(H_{30} > \tau \mid \mathcal{J})$ and minimizes the conditional losses $c_{FN}p$, $1 - p$ and c_A for cool, hot and unknown answers. Probability fitting and Platt calibration use disjoint groups. Constant hot, cool and unknown answers and random rejection at matched retention provide further deployable references. At retention a, expected random-rejection loss is $aL_{\text{all}} + (1-a)c_A$; its counts are expectations. These opened-data comparisons test whether magnitude screening also serves the classification loss.

A refused magnitude can retain an identified hot label. Let L be the largest mean of a *fully observed* 30-day

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window, or minus infinity if none exists. Since $L \leq H_{30}(y)$, L above threshold establishes hot conditions against the complete recorded reference. An 80%-observed mean is not a bound. This secondary rescue leaves magnitude admission unchanged. Observation margins of 0.1 and 0.2 °C are scenarios, not certified sensor bounds; L below threshold never establishes cool conditions.

3.6. Failure diagnosis and robustness

Information-loss experiments remove donors on identical masks and recompute reconstruction and uncertainty. Fitting and calibration information are crossed; this does not identify all causes of international shift. Every accepted Swiss failure is traced by station, year, gap and error sign. Flow and hidden-day proxy residuals explain evaluation outcomes only and do not enter the original frozen screen. Common-population contrasts and truth-assisted nested gaps hold the true 30-day window hidden while extending left, right or symmetrically, retaining boundary distances and infeasibility.

Further assays examine whether visible season and hydrological state can describe slowly varying bias and short residual dependence. They compare a slow-plus-short conditional residual model, 3/5/10-year histories on identical feasible target years, boundary and fragmented gaps, and complete reconstruction–admission–decision systems. Each reconstruction learns and calibrates its own error event; flow-enhanced inputs form a separate information tier. Regional forward-event holdouts remove the target provider's connected components, fit and calibrate global scores through 2014, and evaluate 2015/2018/2022 using strictly earlier local histories. Provider remains a regional proxy. Joint learning/calibration/evaluation budget experiments combine bounded paired-gain evidence (Hoeffding, 1963) with risk evidence; their design, the development assays and a separate future-study specification are reported in SI.

4. Results

4.1. Equal coverage losses have different thermal consequences

Gap position changes H30 error even when missing duration is held fixed. Moving a 30-day gap from zero to full climatological-peak overlap raises climatology-based absolute error by 0.62 °C [0.58, 0.66] in documented daily means (Figure 1a). Broader provider coverage and controls for missing warm-season days retain this position effect. Its magnitude varies with reconstruction and auxiliary information (SI Tables S4–S6).

Length retains an effect. Where the true peak is entirely hidden, extending 30 to 90 days raises climatology error by 0.13 °C [0.10, 0.17], and interpolation error by 1.24 °C [1.12, 1.35]. This exact-hidden length contrast uses 396 distinct station-years in 98 components, versus 4,720 station-years for the position contrast. Their magnitudes cannot rank position against length across different populations. Supplementary common-population and nested-gap results retain extension direction and boundary distance (SI). Fragmentation changes true support exposure despite equal total warm missing days. Target-aware assessment is warranted; one overlap number cannot replace completeness information.

4.2. Window switching often offsets cold peak-window errors

H30 responds to both error within the true peak and the opportunity for another window to replace it. The decomposition holds to numerical precision, with a nonnegative switch increment in every finite case (SI Table S7). Yet 60.4% of positive climatology increments still end below the reference H30. Switching therefore often compensates underestimation of the true peak (Figure 2a).

Climatology oracle mean absolute errors are 0.450 °C originally, 0.398 after permutation, 0.215 after centering and 0.071 after both. The interaction is -0.091 °C, so improvement percentages are not additive mechanism shares. The visible-information correction improves policy MAE by only 0.014 °C [0.008, 0.019]. The oracle diagnoses structure without demonstrating a large available correction.

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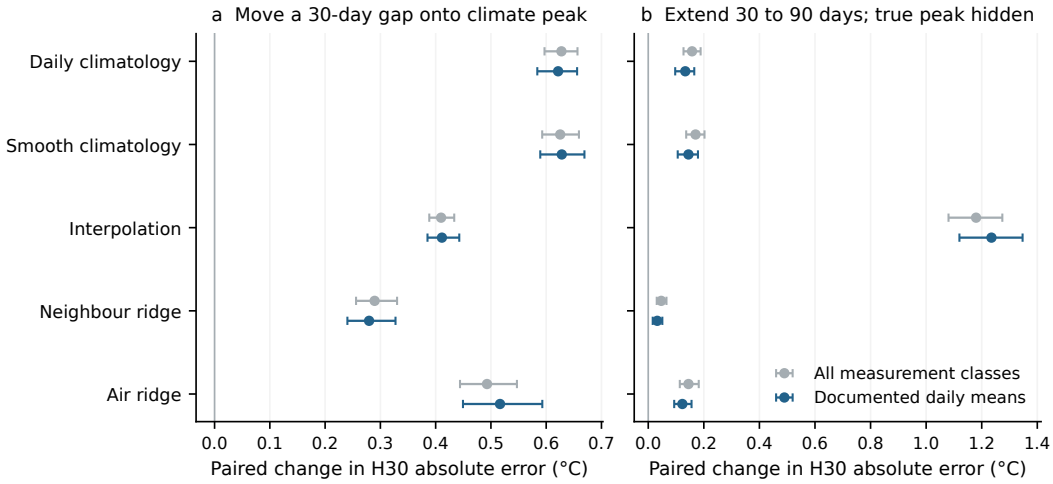


Figure 1: Gap position and duration affect H30 through different contrasts. (a) Change in absolute H30 error when equal 30-day gaps move from zero to full climatological-peak overlap. (b) Change when gaps extend from 30 to 90 days while the true peak remains entirely hidden. Blue denotes documented daily means and gray all measurement classes. Bars give 95% component-bootstrap intervals; starts are averaged within station-years. The documented-mean position comparison covers 4,720 station-years; the exact-hidden duration comparison covers 396 station-years in 98 components.

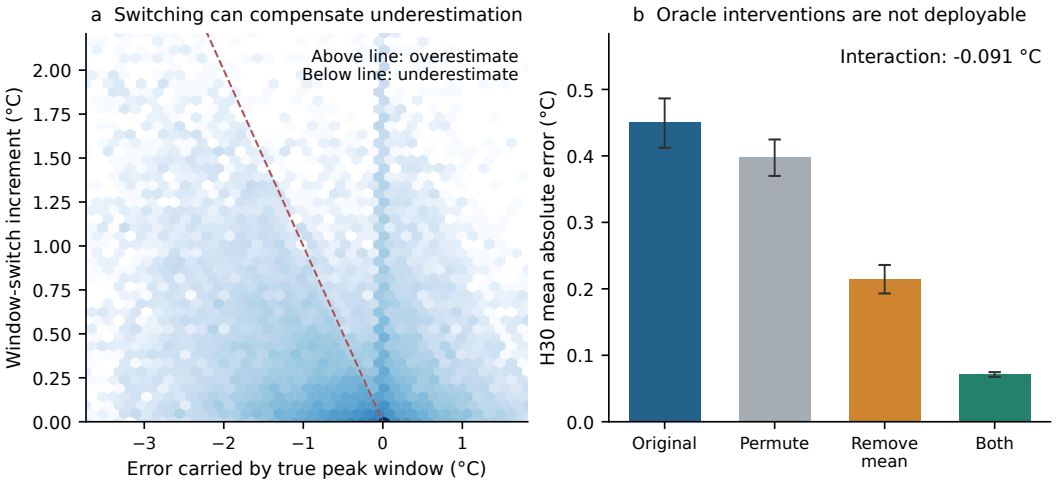


Figure 2: Error propagation through the hottest 30-day statistic. (a) True-window reconstruction error against the nonnegative switch increment for climatology in documented-mean, replicate-zero controlled masks. The diagonal separates net H30 underestimation and overestimation; axes show the central 99% range. (b) Effects of permutation and removal of the hidden residual mean, with 95% component-bootstrap intervals. Both interventions require hidden reference values and have a nonzero interaction. The visible-information correction is evaluated separately in the text.

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4.3. Target diagnostics add information beyond daily accuracy

Diagnosing the intended statistic improves screening at matched reconstruction and learning capacity. Replacing daily disagreement with H30 disagreement increases source retention by 3.8–6.2 percentage points. On independent Swiss admission records after local recalibration, the increase is 9.4 points [6.1, 13.1]. This positive result concerns the diagnostic target.

Additional window summaries provide a smaller and less stable increment. On the admission mixture, uncertain-window source retention exceeds the matched H30-generic score by 1.7–3.8 points across seeds, below the predefined five-point practical margin (Figure 3). Earlier-years-only local inputs give 4.1–7.6 points, but do not constitute a global forward pipeline. Full-procedure intervals remain broad: 2.6 points [-1.9, 8.0] retrospectively and 2.9 [-5.7, 11.3] in the calendar-time forward experiment.

Direct H30 uncertainty gives essentially the same conditional source retention as the larger window representation. Their conditional paired intervals lie within the predefined two-point equivalence band, whereas whole-procedure intervals do not establish equivalence. Complete historical diagnostics are also competitive. This distinguishes target-specific diagnostics from a necessary uncertain-window representation, not models with and without all peak information. Small-model support features, flexible seasonal alternatives, daily-versus-H30 disagreement and component removals appear together (SI Tables S9–S13), rather than selecting a favorable ablation.

The value of explicit overlap also depends on the comparator. Its increment is 3.4–9.5 points for a small logistic model but -1.2 to 1.1 points for a flexible seasonal spline. H1 therefore supports target-specific diagnosis without establishing that one explicit support variable is necessary.

The additional objective comparisons use opened data and do not constitute a new independent test. The choice of learning objective does not reveal a stable extra-window gain in the source cohort. The matched four-cell retrospective analysis contains 54,837 cases in 144 components. Added windows change retention by -0.68 to 0.07 percentage points with quantile learning and 0.02 to 0.12 with direct failure-probability learning. The learning objective does not recover a stable five-point representation advantage. Earlier-local-input and full-procedure results, including their uncertainty, are retained in SI; no non-significant result is declared proof of equivalence.

Across 1,000 full-procedure refits, retrospective window increments average 0.35 points [-4.80, 5.43] for quantile learning and 0.70 [-4.70, 5.82] for failure-probability learning. The wide intervals permit both harm and practically relevant benefit; superiority and equivalence remain unestablished.

A separate controlled assay tests whether stronger probability reconstruction changes this interpretation. It retains all 945 original Swiss controlled candidates and supplies the same developed conditional H30 distribution to the priority, Air2stream-3 and neural points. These additional masks use the existing target histories.

The selected model returns finite distributions for 945 of 945 controlled Swiss test candidates. Among these predictions, H30 CRPS is 0.239 °C, nominal-90% coverage is 81.0%, and mean band width is 0.931 °C. Warm-year signed bias is -0.334 °C; coverage on the 124 majority-low-flow gaps with complete flow records is 71.0%. These are overlapping diagnostic conditions, not identified causal effects. The mean absolute functional-point change from eight to thirty-two draws is 0.091 °C. Sampling stability cannot repair distributional miscalibration.

The window increment depends on the delivered point and direct comparator. For the neural point, the window head accepts 446 candidates with 16 failures, versus 370 with 6 for the boosted direct head: a conditional 8.0-point retention increase. However, the native CDF rule already accepts 474 with 12 failures, exceeding the window head's retention with lower sample risk. For Air2stream-3, boosted-head acceptance instead changes from 404 to 380, with failures changing from 11 to 7. No cell has positive formal admission. These opened-data operating points neither establish a new independent advantage over the strongest direct reference nor support a universal negative claim about window information.

The trajectory scores are not calibrated probabilities. Raw 90% scenario-band coverage is only 71.6% in retrospective Swiss admission and 60.7% prospectively. Their representation comparison does not establish that a better joint predictive distribution could never make window information useful.

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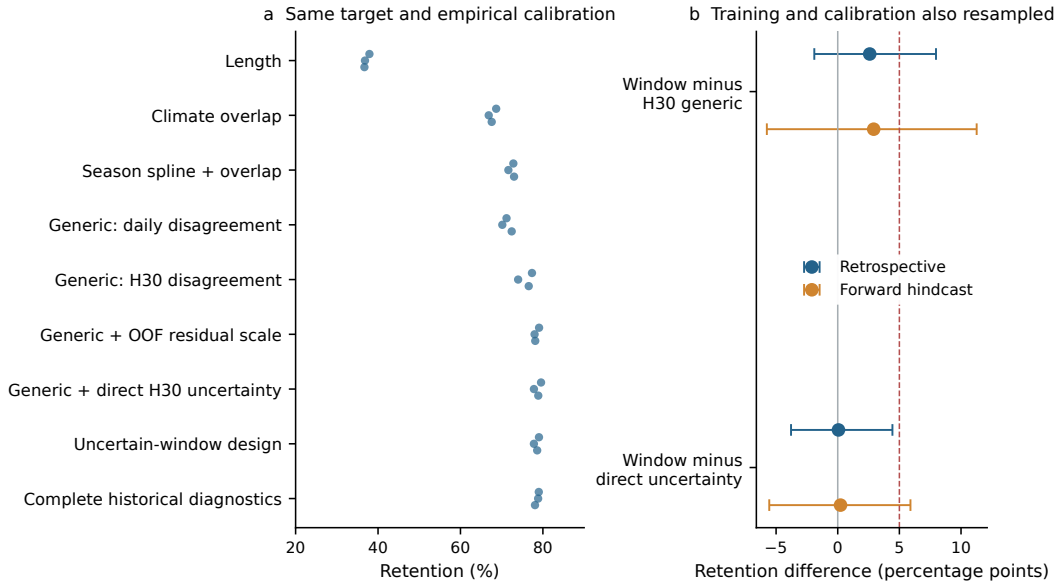


Figure 3: The diagnostic target and incremental window representation are evaluated separately. (a) Retention under the common empirical 5% calibration target in the documented-mean source admission cohort. Each dot represents one of three component-split sensitivities. OOF denotes out-of-fold residuals of the actual reconstructor. (b) Whole-procedure intervals refit learning and calibration and resample evaluation components. The dashed line marks the predefined five-point practical margin. Conditional similarity does not establish full-procedure equivalence.

4.4. Calibration shift concentrates failures across river conditions

The source-calibrated threshold does not retain its empirical risk target after unchanged transfer. On 669 Swiss documented-mean admission cases from 7 held-out components, F0 retains 56.1% and incurs 8.8% failures, against the 5% calibration target (Figure 4). F1 local recalibration changes these values to 47.5% and 4.1%; F2 local fitting and calibration give 44.4% and 3.0%. The seven-component intervals establish neither population risk above 5% for F0 nor a population guarantee below 5% for F1 or F2.

The 669 retrospective Swiss masks represent 105 distinct station-years in seven stations, with 17 distinct hot station-years; they are generated single gaps retaining annual boundary observations. Sixteen of 33 frozen admitted failures occur at CH_2327, Dischmabach–Davos, Kriegsmatte, and all sixteen underestimate H30. Fifteen use air reconstruction. The official station inventory confirms a roughly 43-km^2 alpine catchment with 0.7% glacial area (Federal Office for the Environment, 2026); this is not evidence of a specific regulation or groundwater class. Gap-period cold proxy errors can exceed visible-period errors, while low-flow overlap varies. Source donor removal raises unscreened error but does not reproduce the full Swiss distribution. Persistent peak-season proxy mismatch is a testable explanation, not an identified low-flow or melt mechanism (Michel et al., 2020). All original groups remain in the primary result.

The calendar-time forward Swiss pool is smaller, 173 cases. Unchanged-transfer retention and risk are 47.4% and 9.8%; recalibration gives 37.0% and 1.6%. French aggregates, unseen-provider and intercontinental transfers, joint certificate delivery and complete diagnostics are separate (SI Table S14). They do not supply the missing independent five-point, common-risk representation advantage. H2 is not supported.

Reconstruction remains consequential. Across both new measurement scopes on matched forcing-supported controlled cases, Air2stream-3 has MAE 0.254 °C versus 0.280 for the fixed policy. Air2stream-8 has 0.203 versus 0.342 on 821 cases, but additionally uses discharge. The original frozen reference experiment did not substitute process points into risk training. A supplementary factorial now refits each reconstruction's own error head, calibration and use decision on matched Swiss cases (SI), without changing that original comparison. The priority point is a controlled comparison object, not the best

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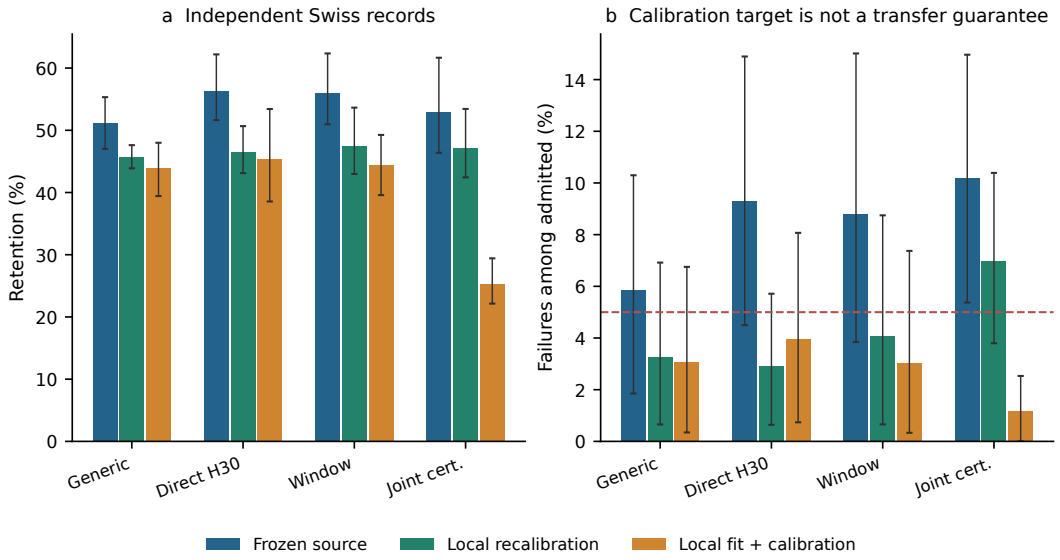


Figure 4: Transfer changes the relation between retention and observed failure. The Swiss test pool contains 669 admission cases in seven components. F0 preserves the source model and threshold; F1 recalibrates the same model on target calibration stations; F2 fits on new development stations and calibrates locally. The joint certificate selects its own reconstruction; other arms deliver the common point. Bars show descriptive component-bootstrap uncertainty. The dashed 5% line is the empirical calibration target.

attainable reconstruction under every input budget.

Calibration-set composition also matters. With the developed model and heads held fixed, all 247 subsets of three, five, seven or nine target calibration groups yield 741 score/subset evaluations on the same 945 tests. With three groups, the native CDF rule has empty admission in 28 of 84 subsets; its nonempty test risks range from 1.0% to 24.4%. Even seven-group subsets reach a sample risk of 6.2%. Every formal calibration remains empty. These ranges measure conditional sensitivity to already opened calibration groups, not a powered independent test; no subset is selected to improve the result.

The formal selector returns no positive admission under the available evidence. Constructed rare-group and correlated-failure experiments show why the empirical selector need not be conservative (SI Table S18). Descriptive hydrological, anomaly and event strata expose variation hidden by pooled percentages (SI Tables S15–S17). Nondegenerate cofailure stress is confined to held-out opened-source networks (SI Figure S2).

4.5. Numerical screening and classification have different benefits

Returning more sufficiently accurate H30 magnitudes does not necessarily minimize hot-year classification loss. In the Swiss admission cohort, F0 refuses 46.4% of true hot cases, 43.4% of cool cases and 43.9% overall. Refusal changes the evaluated population as well as the number of available labels (Figure 5; SI Table S24).

At missed-hot cost five and unknown-answer cost 0.05 relative to a false-hot answer, loss changes by -0.3368 per original candidate [-0.4059, -0.2720] versus reconstructing all. This is not sufficient to justify the selective policy: its absolute loss is 0.1191, whereas declaring every record unknown costs only 0.05. We added this deterministic comparator in a post-test audit without changing predictions or the predefined cost grid (SI Table S23). Some other declared cells beat both fixed alternatives at their point estimates, not under a simultaneous or measured management-utility guarantee. With inexpensive errors and costly abstention, even the reconstruct-all comparison can reverse. Observed-hot rescue adds only 2 labels here and 1 prospectively. H3 holds only for its specified reconstruct-all scenario comparison, not a generally superior operational policy.

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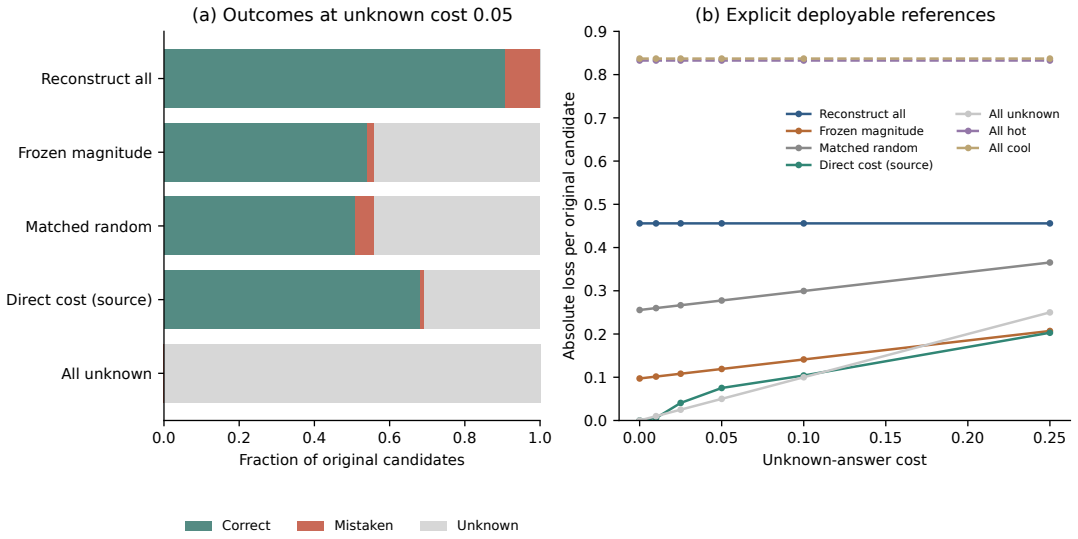


Figure 5: Decision value depends on the output and declared costs. All results retain the same 669 Swiss admission candidates and missed-hot cost five. (a) Correct, mistaken and unknown fractions at unknown-answer cost 0.05. Matched random rejection uses expected counts at the frozen screen's retention. (b) Absolute losses across unknown-answer costs, including constant answers and the direct-cost probability reference. The added probability policy uses opened data; the original frozen screen is unchanged. Costs are experimental rather than measured ecological or agency losses.

At the same Swiss retrospective FN5/A0.05 setting, random rejection at the original screen's retention has expected loss 0.2775. Direct hot-probability decisions using source-calibrated direct features have loss 0.0752, still above always unknown at 0.0500; their forward loss is 0.0289. Thus neither a magnitude tolerance nor a favorable reconstruct-all comparison determines the best classification action. These added policies use opened data and establish no new confirmatory superiority.

At missed-hot cost five and unknown cost 0.05 on the same 945-candidate pool, the neural direct-head magnitude screen has loss 0.0304, versus 0.0634 for its window head. The calibrated CDF cost decision has loss 0.0530; all-unknown loss is 0.0500. Numerical retention, numerical risk and label loss therefore rank these choices differently. These are secondary declared-cost comparisons, not ecological losses or new confirmatory wins.

Recorded-calendar replay further limits inference from favorable constructed gaps. The eligible new calendars come from French project aggregates. Their 168 masks in 10 components incur 21.4% unscreened failures, compared with 26.6% predicted by the source-fitted composition rule (SI Table S20). These are replay risks conditional on the selected geometries, rather than natural failure prevalence across the monitoring archive.

5. Discussion

5.1. Thermal-statistic fitness requires a target-specific diagnosis

The central finding is that the target of diagnosis matters more consistently than the addition of an explicit peak-window representation. H30 diagnostics retain more usable estimates than daily diagnostics under matched comparisons, whereas added window summaries do not establish the predefined practical increment over strong direct references. This extends earlier evidence on missing-data timing and seasonal thermal-peak reconstruction (Johnson et al., 2021; Beaufort et al., 2022) by separating the diagnostic target from its representation. Seasonal support is already known to matter; the additional contribution is the comparison of information needed to assess the derived statistic.

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The contrast has a statistical explanation. Daily errors summarize the series, while H30 accumulates errors over a selected warm period and then maximizes over possible periods. A sustained offset can therefore matter more than dispersed daily deviations with a similar aggregate error. Once a complete conditional H30 distribution is available for a fixed delivered point, it determines the probability of exceeding tolerance. Window summaries may improve estimation of that distribution, particularly when the distribution model misses relevant state or dependence. Their benefit must be demonstrated against the direct target-risk reference.

The error decomposition also qualifies a common interpretation of peak reconstruction. A positive switch increment can compensate cold error within the true peak, so its sign alone does not identify a falsely hot reconstructed extreme. The strong oracle response to residual centering indicates error structure that affects the statistic, but the small visible-information correction shows that this structure is not yet equally recoverable in practice. The intervention interaction and conditioning on a true maximum further limit physical attribution from the decomposition.

5.2. Calibration transport depends on the deployment information

The new-station experiments distinguish informative ranking from transferable calibration. A score can order source cases usefully while its threshold admits too many target failures. In the present transfer, station conditions and reconstruction inputs change together: source reconstructions often use neighbors, whereas Swiss reconstructions rely mainly on air temperature. Source donor removal isolates part of that information change, but does not reproduce the full target error distribution. Consequently, the international contrast cannot be assigned to a single hydrological cause.

The concentrated cold failures at CH_2327 are consistent with a peak-season air-proxy mismatch and motivate tests of conditional bias and multiweek persistence. They do not establish whether low flow, melt inputs or another process produced that mismatch. This distinction complements evidence that similar in-distribution river-temperature accuracy need not imply similar behavior under environmental shift (Topp et al., 2023). Predictive generalization and transport of a calibrated admission threshold require related but separate tests.

Local calibration improves the observed operating point, but its interpretation depends on the number and composition of independent examples. Record-overlap components prevent duplication without ensuring independent catchment climates or heat events. The formal rule requires deployment-representative components and controls a group-expectation ratio. Empty formal admission indicates insufficient evidence for that rule, rather than universal record unusability. The calibration-subset results illustrate why additional masks cannot substitute for additional independent calibration units.

5.3. Archive value depends on the requested output

An archive user may require an H30 magnitude, a hot-year label or a regional frequency. Each output has a different relation to reconstruction error and refusal. A complete observed window can establish a hot label after magnitude refusal, although few labels were recovered here. A below-threshold observed window cannot establish a cool year. At regional scale, selective accuracy must be considered together with unknown fractions, hot/cool refusal and original-pool frequency bounds.

The different policy rankings follow from these different objectives. A numerical tolerance constrains magnitude error, whereas hot-year loss depends on threshold crossing and the relative costs of missed, false and unknown answers. The experimental cost grid exposes that distinction without estimating ecological damage, field-revisit expense or agency preferences. Similarly, the finite-history anomaly threshold describes local thermal conditions rather than a biological standard.

A reduction relative to reconstructing all can coexist with higher loss than a constant unknown answer or another directly optimized policy. The stronger references therefore change the interpretation of the original H3 comparison. An application that requires a minimum information return must specify that requirement

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before evaluation. Likewise, adaptive archive retrieval is meaningful only when an actual record, feasible retrieval action and measured cost exist. Freely available backup and forcing data should be incorporated into the information set.

5.4. Evidence boundaries and the next independent comparison

The experiments require complete reference and training years, limiting their representation of intermittent and persistently gapped archives. The new station histories are European mountain catchments with few independent calibration/test components and no within-bundle donor variation. Available glacier and snow attributes do not supply verified regulation or groundwater classes. Measurement definitions, quality flags and project aggregation further constrain comparisons. Dry channels require particular care because absent water can change the definition of a temperature estimand.

Constructed gaps identify error responses to prescribed missingness, while calendar replay conditions on recorded gap shapes. Neither supplies the unknown temperatures of the original naturally incomplete years or identifies the causes of instrument failure. Derived and published thermal-event definitions also differ, and incomplete discharge leaves some compound conditions unknown. Current-vintage forward hindcasts restrict information to earlier calendar years without recovering the data vintages that would have been available in real time.

The developed probability comparison strengthens the direct references while retaining a finite model budget. Its neural window head improves retention relative to the matched boosted direct head, but the native CDF rule already has higher retention and lower sample risk. This positive within-head increment remains relevant; it establishes neither a win over the strongest direct reference nor a universal negative conclusion about window information. Undercoverage and conditional cold bias likewise constrain interpretation of the scenario distributions. The source-law joint-budget experiment supports no 80% design in its evaluated grid and supplies no recruitment target for a new neural candidate or climate (SI).

Independent-backup qualification remains a substantive evidence gap. No DFO pair-year meets the required complete-history support (Fisheries and Oceans Canada, 2024). The two separately evaluated short-history records contain cold-season partial daily deficits and retain their observed hot windows, making zero H30 error weak evidence for difficult recovery. A further metadata-led search acquired paired channels at seven USGS stream sites, but none of 25 candidate pair-years passed all reference requirements (SI). These exclusions preserve the original support and sensor-consistency criteria.

The present evidence therefore supports a study of target diagnosis and failure boundaries. A new representation requires a separate independent comparison in which predictive choices, calibration resources, deployment units and intended use are fixed before outcomes. Joint success must combine risk evidence with a practical gain over both native-CDF and matched-capacity direct-risk references. Additional masks, a favorable subgroup or an oracle intervention cannot replace that comparison.

6. Conclusions

Using 6,170 complete reference station-years and paired gap experiments, we show that thermal-statistic fitness depends on both missing-data geometry and the target of diagnosis. Equal 30-day gaps differ in climatology-based H30 error by 0.62 °C when moved into the climatological peak. H30 diagnostics increase locally recalibrated Swiss retention by 9.4 points relative to daily diagnostics. Additional peak-window summaries do not establish the predefined five-point advantage over strong direct target-risk references.

The original source-calibrated rule incurs 8.8% failures against a 5% target on 669 Swiss cases; local recalibration lowers observed risk to 4.1%, with population reliability unresolved by the small independent cohort. Classification loss can favor a different action from numerical screening. Assessing an incomplete thermal archive consequently requires separate evidence for the reconstructed statistic, its calibrated admission rule and the requested output.

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Open Research Statement

Public products are reused under provider terms. Existing rights and measurement definitions are in `DATA_RIGHTS.md`. New contracts, acquisition records, event/forcing resources and identity exclusions are under `results/wrr_submission_ready/major_revision_20261020c/`. The mountain resource has DOI `10.4211/hs.7a45ee06851e41d99e4affb7110d5f97`; continuous temperatures were reacquired, not inferred from extreme values. Local availability is not permission to redistribute every raw series.

Scripts 348–373 implement the original experiments and evidence, with method and deployment locks predating test scoring. Scripts 374–393 implement the supplementary objective, decision, information, trajectory, developed-probability, joint-planning and calibration-budget analyses under `results/wrr_submission_ready/followup_20261021/`. Generators 383, 391 and 393 bind numerical text, tables and figures to their result files. Scripts 394–398 document the additional public backup-source search, including native approval flags and all candidate exclusions. Public-source attribution and redistribution scope are recorded in `DATA_RIGHTS.md`. Original protocols and the separate future-study specification remain available in the accompanying repository; a permanent archival deposit is required before submission.

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Your acknowledgements will appear here.

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